The three Dirichlet boundary conditions

Exact initial solution with its boundary values, and g unrolled along the perimeter over time.

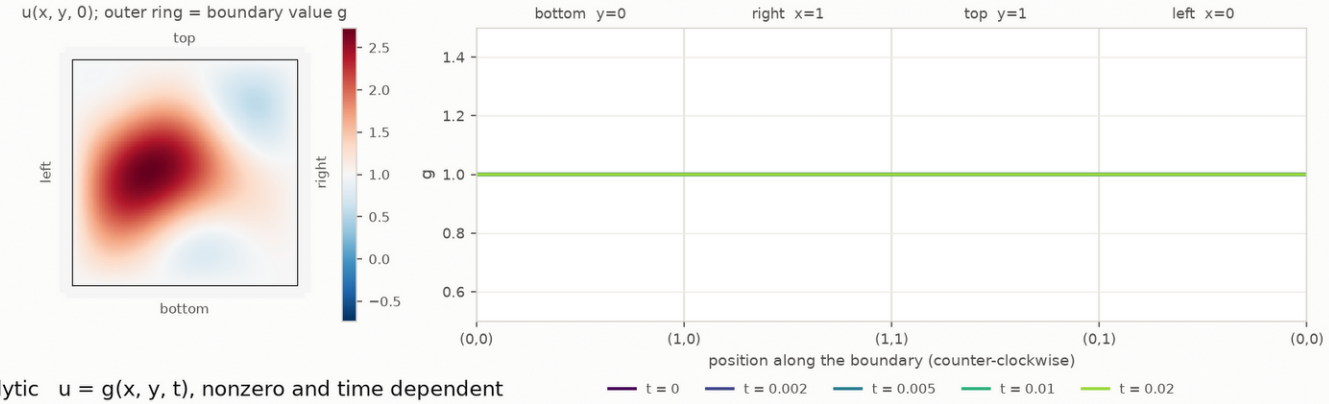
Boundary conditions used in the error analysis

Same four modes in every case; wavenumbers are multiples of pi. Zero phases make every mode vanish on the boundary; the analytic case turns the phases on.

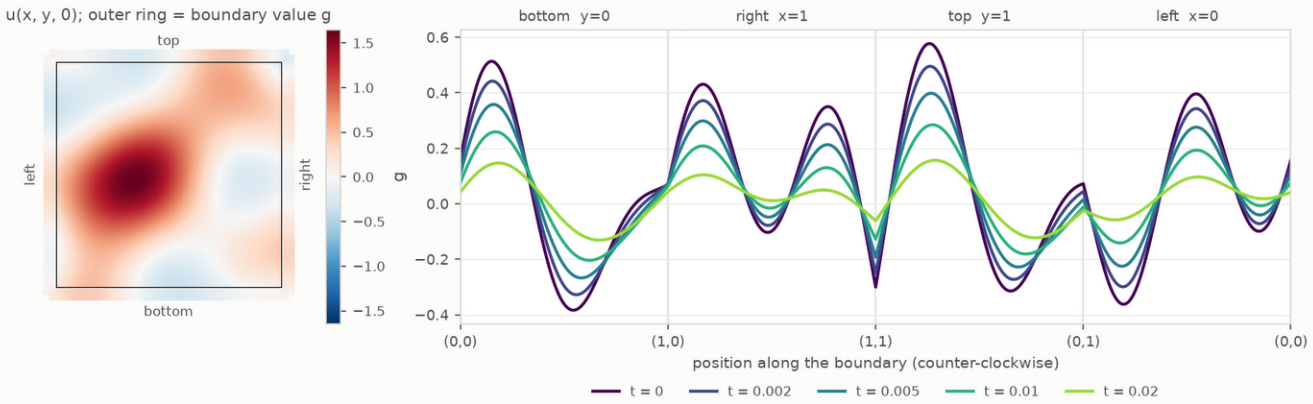
1. homogeneous u = 0 on the boundary (default)



2. constant u = 1 on the boundary

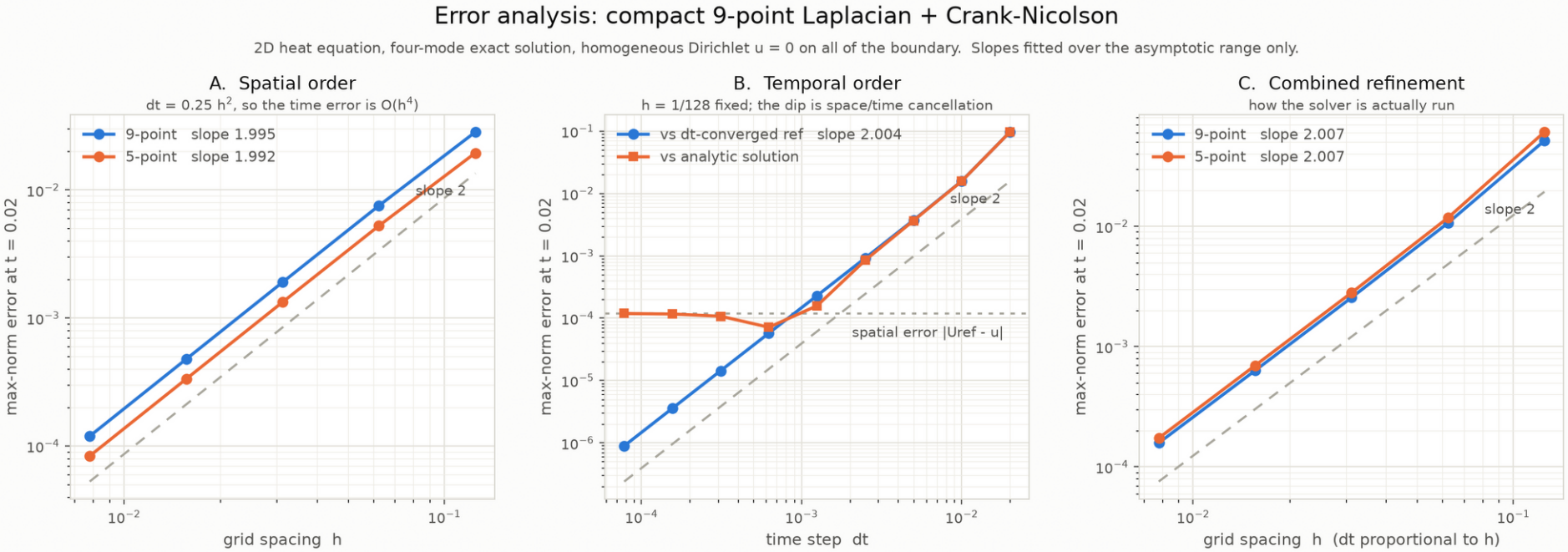


3. analytic u = g(x, y, t), nonzero and time dependent



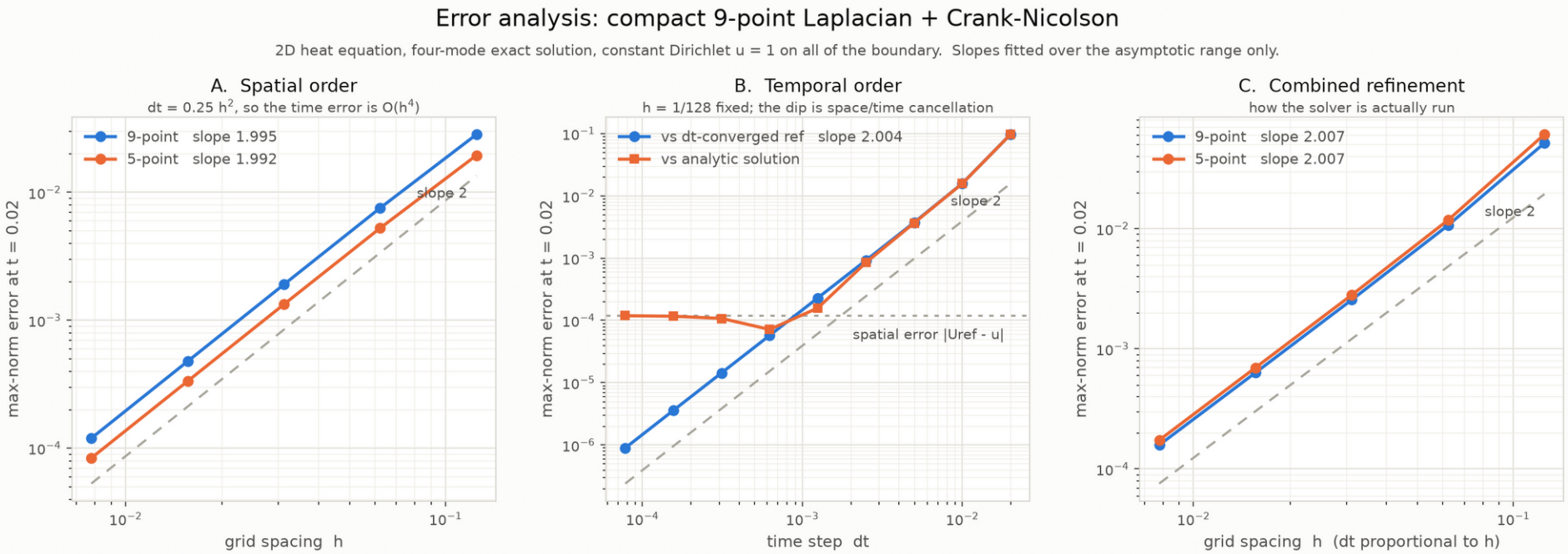
Error analysis: homogeneous boundary (u = 0)

Default case. Each mode is an exact grid eigenvector, so the error is pure eigenvalue error.



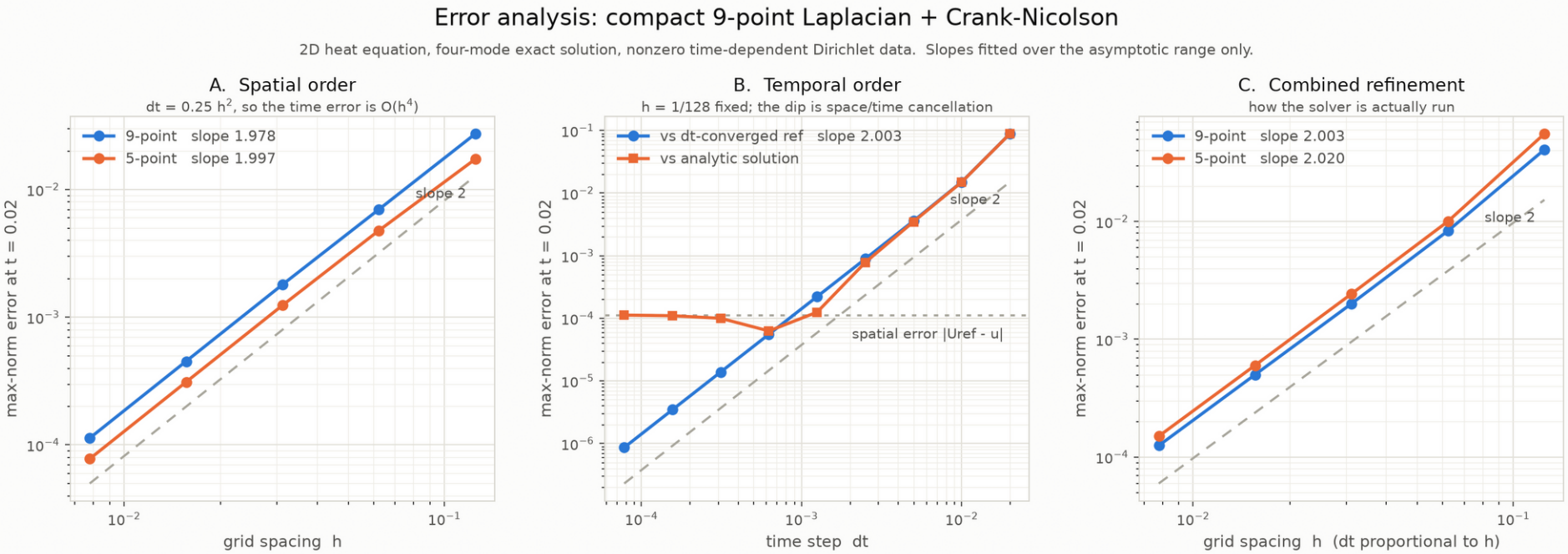
Error analysis: constant boundary (u = 1)

Matches the homogeneous case to within roundoff; a consistency check, not a new accuracy test.



Error analysis: time-dependent boundary data

The most demanding case: nonzero, time-dependent g exercises the boundary vector b(t).



Isotropy: 9-point vs 5-point stencil

A diffused radially symmetric blob; the 9-point stencil's isotropic error keeps it round.

Diffusing a narrow Gaussian: 5-point vs compact 9-point

Same grid (h = 1/96), same time step, same final width. Dashed ring: half-max radius on the x-axis.
Both panels in each row share one colour scale.

